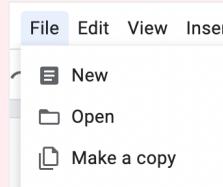


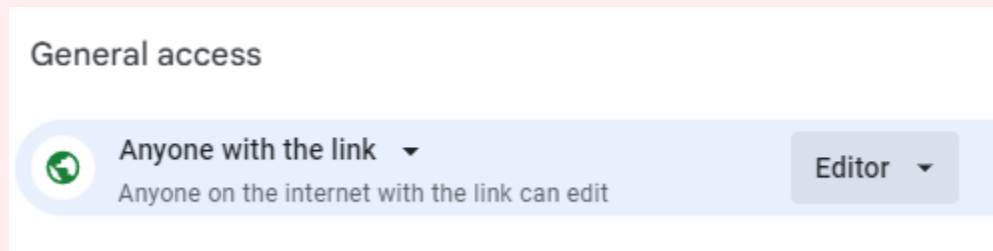
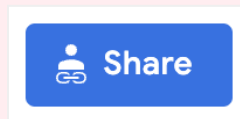
Instructions for Copying and Sharing this Document

 **DELETE THIS BOX BEFORE SUBMITTING!!**

Step 1: **Click** “File -> Make a Copy” to make a copy of this document that you can edit.



Step 2: **Change** the Share settings to “Anyone with Link -> Editor”. This will allow our graders to leave comments on your submission.



CYB102 Project 5

( [Instructions Page](#))

 Student Name: Giancarlo Montes

 Student Email: giancarlo.montes001@Mymdc.net

Reflection (Required)

 **Reflection Question #1:** If I had to **explain “what is SIEM” in 3 emojis**, they would be...
(Feel free to put other comments about your experience in this unit here, too!)



 **Reflection Question #2:** What field do you think is most important for logs to have?

I believe the (Time) timestamp field is the most important for logs to have, because without knowing the proper time frame it will be difficult to find certain logs.

 **Shoutouts:** Share appreciation for anyone who helped you out with this project or made your day a little better!

Group #17

CTF Challenges (Required)

Use the answer boxes below to document any CTF challenges you completed. If you don't complete a particular challenge, leave it blank.

Part 1 - Searching the Netflix Data (1pt each)

index=main source=Netflix

 **Challenge 1:** How many TV shows on Netflix are in the Docuseries genre?

Solution:

1. **170** TV shows on Netflix are in the Docuseries genre.

```
1 index=main host=Netflix type ="TV Show" listed_in="Docuseries" | stats count
```

✓ 170 events (before 10/17/24 7:07:59.000 PM) No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

20 Per Page ▼ ✎ Format Preview ▼

count ↕

170

Challenge 2: How many movies on Netflix have a rating of TV-PG?

Solution:

1. **1080** movies on Netflix has a rating of TV-PG.

2. **index=main host=Netflix type ="Movie" rating="TV-PG" | stats count**

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Dashboards

New Search

```
1 index=main host=Netflix type ="Movie" rating="TV-PG" | stats count
```

✓ 1,080 events (before 10/17/24 7:24:58.000 PM) No Event Sampling ▾

Events Patterns **Statistics (1)** Visualization

20 Per Page ▾ ✎ Format Preview ▾

count ↕
1080

 **Challenge 3:** How many movies on Netflix were released in the year 2020?

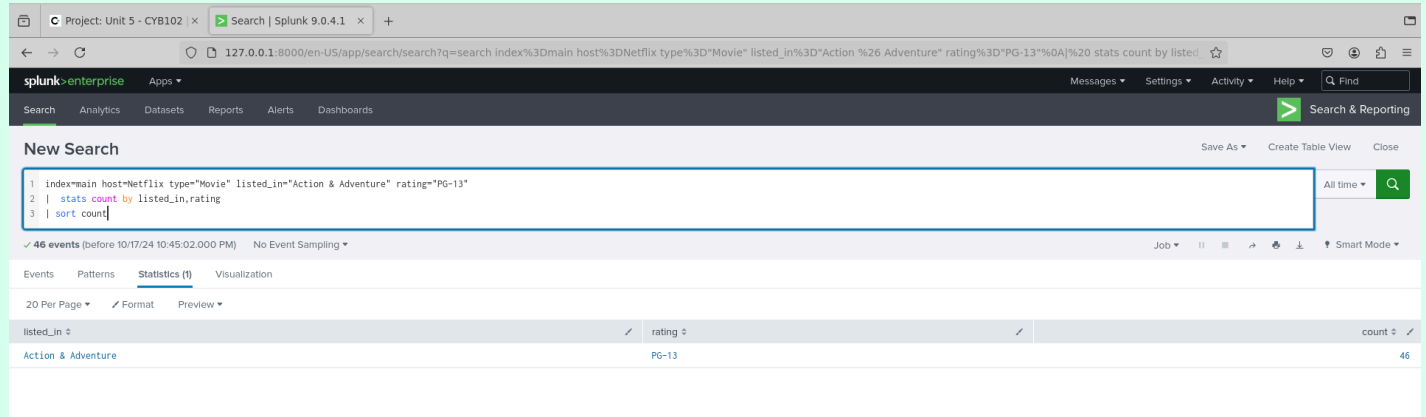
Solution:

1. **1034** movies were released in the year 2020 on Netflix.
2. **index=main host=Netflix type ="Movie" release_year="2020" | stats count**

Challenge 5: How many movies on Netflix are listed as action and are rated PG-13?

Solution:

1. **46** movies are listed as action and are rated PG-13.
2. **index=main host=Netflix type="Movie" listed_in "Action & Adventure" rating="PG-13"**
| stats count by listed_in_rating
| sort count



The screenshot shows the Splunk Search interface. The search bar contains the query: `1 index=main host=Netflix type="Movie" listed_in="Action & Adventure" rating="PG-13"`
`2 | stats count by listed_in_rating`
`3 | sort count`

The results show 46 events. The table displays the following data:

listed_in	rating	count
Action & Adventure	PG-13	46

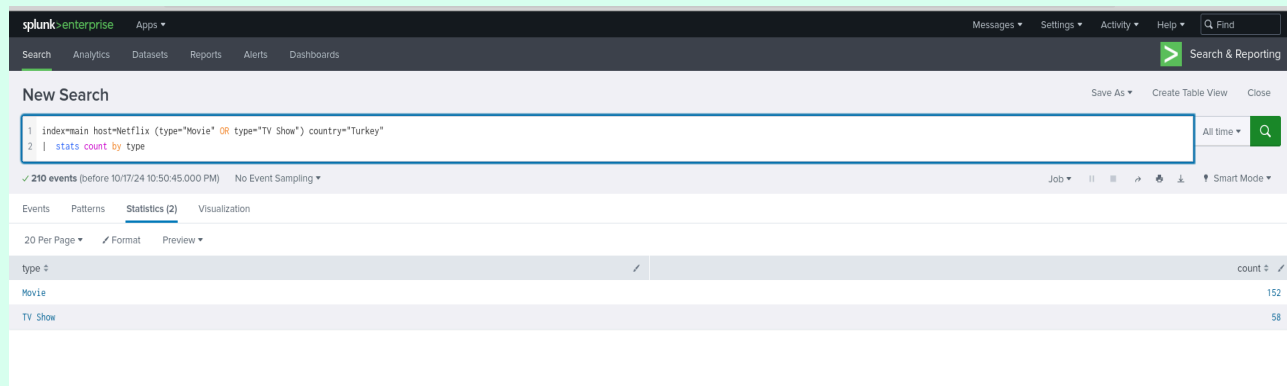
Challenge 6: How many movies and TV shows on Netflix have their country of origin as Turkey?

Solution:

1. **210** Movies and TV shows on Netflix have their country of origin as Turkey.

We get this answer by adding (Movies: 152) + (TV Shows: 58) = 210

2. **index=main host="Netflix" (type="Movie" OR type="TV show") country="Turkey"**
| stats count by type



The screenshot shows the Splunk Search interface. The search bar contains the query: `1 index=main host=Netflix (type="Movie" OR type="TV show") country="Turkey"`
`2 | stats count by type`

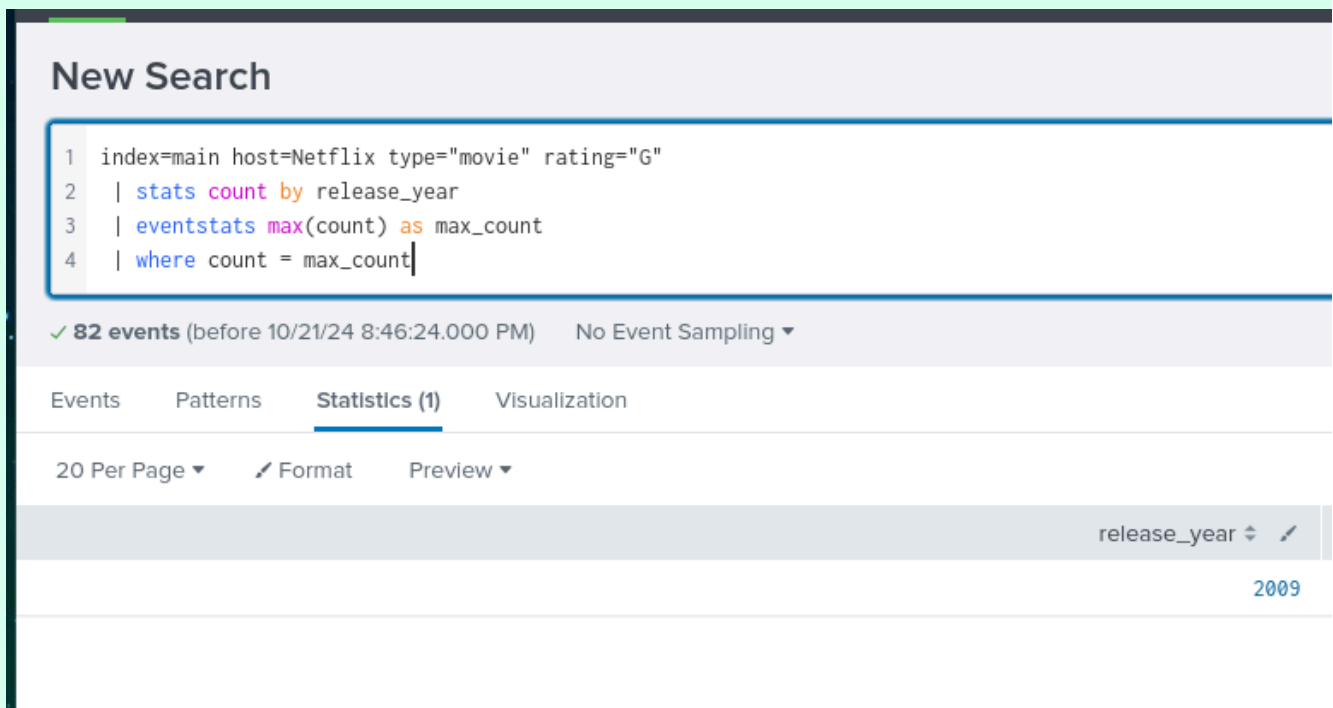
The results show 210 events. The table displays the following data:

type	count
Movie	152
TV show	58

Challenge 7: Which release year had the most movies rated G? (Not TV-G)

Solution:

1. The year: **2009** had the most movies rated G.
2. **index=main host=Netflix type="movie" rating="G"**
| stats count by release_year
| eventstats max(count) as max_count
| where count = max_count



New Search

```
1 index=main host=Netflix type="movie" rating="G"
2 | stats count by release_year
3 | eventstats max(count) as max_count
4 | where count = max_count
```

✓ 82 events (before 10/21/24 8:46:24.000 PM) No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

20 Per Page ▼ Format Preview ▼

release_year
2009

Challenge 8: What two TV-Y7 rated shows were released in 2019 and were added to Netflix on November 22, 2019?

Solution:

1. The two TV-Y7 rated shows released on November 22, 2019 were: **Trolls** and **The Dragon Prince**.

2. **index=main host=Netflix type="TV Show" rating="TV-Y7" release_year="2019"**
date_added="November 22, 2019"
| dedup title
| table title, release_year

New Search

```
1 index=main host=Netflix type="TV Show" rating="TV-Y7" release_year="2019" date_added="November 22, 2019"
2 | dedup title
3 | table title, release_year
```

✓ 2 events (before 10/21/24 8:47:46.000 PM) No Event Sampling ▼

Events Patterns **Statistics (2)** Visualization

20 Per Page ▼ / Format Preview ▼

title ↕

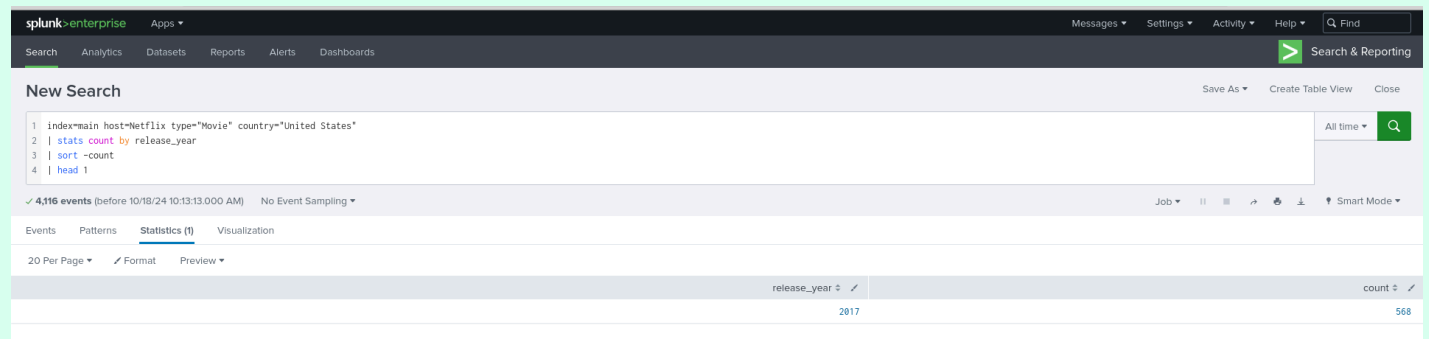
Trolls: The Beat Goes On!

The Dragon Prince

Challenge 9: Which year had the most movies from the United States?

Solution:

1. **2017** had the most movies from the United States with a total of (568)
2. **index=main host=Netflix type="Movie" country="United States"**
| stats count by release_year
| sort -count
| head 1

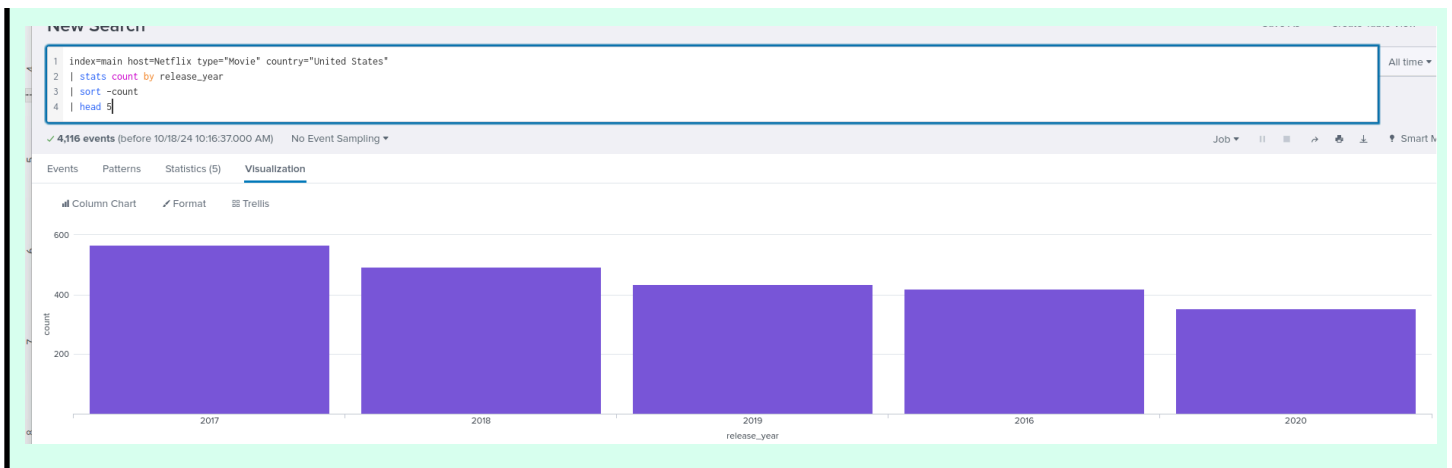


The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items. Below that, a 'New Search' bar contains the following search query:

```
1 index=main host=Netflix type="Movie" country="United States"
2 | stats count by release_year
3 | sort -count
4 | head 1
```

Below the search bar, it indicates '4,116 events (before 10/18/24 10:13:13.000 AM) No Event Sampling'. The results are displayed in a table with two columns: 'release_year' and 'count'. The first row shows '2017' with a count of '568'.

release_year	count
2017	568



Challenge 10: What is the oldest TV show by Release Year on Netflix?

Solution:

1. The oldest TV show by Release Year on Netflix is **Pioneers: First Women Filmmakers***

index=main host="Netflix" type="TV Show"

| sort release_year

| head 1

| table title, release_year

New Search

```
1 index=main host="Netflix" type="TV Show"
2 | sort release_year
3 | head 1
4 | table title, release_year
```

✓ 1 event (before 10/21/24 8:53:48.000 PM) No Event Sampling

Events Patterns **Statistics (1)** Visualization

20 Per Page Format Preview

title

Pioneers: First Women Filmmakers*

release_year ↕ ✎

1925

Part 2 – Investigating the Malware (2pts each)

For Part 2 we are investigating an attacker who got into our systems that happened at PathCode Inc.

For these logs use index=pathcode

Challenge 11: What was the IP address that uploaded the malware (MD5 hash: 3AADB7E527FC1A050E1C97FEA1CBA4D)

Solution:

- 192.168.1.10** is the IP address that uploaded the malware.
- index=pathcode | search "3AADB7E527FC1A050E1C97FEA1CBA4D" "uploaded"**

New Search

1 index=pathcode

2 | search "3AADB7E527FC1A050E1C97FEA1CBA4D" "uploaded"

✓ 2 events (before 10/18/24 10:25:10.000 AM) No Event Sampling ▼

Events (2)

Patterns

Statistics

Visualization

Format Timeline ▼

– Zoom Out

+ Zoom to Selection

× Deselect

List ▼

✎ Format

20 Per Page ▼

< Hide Fields

≡ All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mdav 1

i	Time	Event
>	6/4/2023 17:59:00.000 PM	6/4/2023 17:59:192.168.1.10,File Uploaded,Opera/75.0.3969.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
>	6/4/2023 17:59:00.000 PM	6/4/2023 17:59:192.168.1.10,File Uploaded,Opera/75.0.3969.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv

i	Time	Event
>	6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3969.218,200, host = uploadedhashes source = /home/codepath/Files/Splunk-5-6-7/up
>	6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3969.218,200, host = uploadedhashes source = /home/codepath/Files/Splunk-5-6-7/up

IP

1 Value, 100% of events

Selected

Reports

[Top values](#)
[Top values by time](#)
[Rare values](#)

[Events with this field](#)

Values	Count	%
192.168.1.10	2	100%

Challenge 12: What usernames did that IP address try to login to the system as? Which one did they upload a file as?

Solution:

- The IP address 192.168.1.10 tried to login to the system under the usernames: **(ABurke, Admin and Pi)**
- index=pathcode | search "192.168.1.10"**

1 index=pathcode | search "192.168.1.10"

✓ 26 events (before 10/21/24 7:12:08.000 PM) No Event Sampling ▾

Events (26) Patterns Statistics Visualization

Format Timeline ▾ — Zoom Out + Zoom to Selection × Deselect

List ▾ ↗ Format 20 Per Page ▾

< Hide Fields

≡ All Fields

SELECTED FIELDS

a host 4

a source 4

a sourcetype 1

INTERESTING FIELDS

a action 1

bytes_received 3

bytes_sent 3

a category 4

date_hour 4

date_mday 1

date_minute 10

a date_month 1

a date_wday 1

i	Time	Event
>	6/4/23 3:05:00.000 PM	6/4/2023 15:05,ABurke,192.168.1.10,Login Attempt,Firefox/89 host = webserver02 source = /home/codepath/Files/Splunk-5-6-
>	6/4/23 3:05:00.000 PM	6/4/2023 15:05,ABurke,192.168.1.10,Login Attempt,Firefox/89 host = webserver02 source = /home/codepath/Files/Splunk-5-6-
>	6/4/23 3:04:00.000 PM	6/4/2023 15:04,Pi,192.168.1.10,Login Attempt,Firefox/89.0,2 host = webserver02 source = /home/codepath/Files/Splunk-5-6-
>	6/4/23 3:04:00.000 PM	6/4/2023 15:04,Pi,192.168.1.10,Login Attempt,Firefox/89.0,2 host = webserver02 source = /home/codepath/Files/Splunk-5-6-
>	6/4/23 2:54:00.000 PM	6/4/2023 14:54,Admin,192.168.1.10,Login Attempt,Firefox/89. host = webserver02 source = /home/codepath/Files/Splunk-5-6-
>	6/4/23	6/4/2023 14:54,Admin,192.168.1.10,Login Attempt,Firefox/89.

Username



3 Values, 61.538% of events

Selected

Yes

No

Reports

Top values

Top values by time

Rare values

Events with this field

Values	Count	%	
ABurke	12	75%	
Admin	2	12.5%	
Pi	2	12.5%	

Challenge 13: What was the User Agent String of the attacker when they successfully uploaded a file?

Solution:

1. The user agent string of the attacker when they successfully uploaded a file was:
Opera/75.0.3639.218
2. **index=pathcode host=uploadedhashes OR uploaded OR 192.168.1.10**

New Search

1 index=pathcode host=uploadedhashes OR uploaded OR 192.168.1.10

✓ 44 events (before 10/21/24 9:24:49.000 PM) No Event Sampling ▼

Events (44) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

List ▼ Format 20 Per Page ▼

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS a host 4 a source 4 a sourcetype 1 INTERESTING FIELDS # date_hour 4 # date_mday 1 # date_minute 18 a date_month 1 a date_wday 1 # date_year 1 a date_zone 1 a Event 4		>	6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3639.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
		>	6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3639.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
		>	6/4/23 5:49:00.000 PM	6/4/2023 17:49,192.168.1.8,File Uploaded,Safari/13.1.1,200,Style.css,75441F0A76544B5983E31DFA1C0D8EA2 host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
		>	6/4/23 5:49:00.000 PM	6/4/2023 17:49,192.168.1.8,File Uploaded,Safari/13.1.1,200,Style.css,75441F0A76544B5983E31DFA1C0D8EA2 host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
		>	6/4/23 5:34:00.000 PM	6/4/2023 17:34,192.168.1.4,File Uploaded,Firefox/89.0,200,Homepage.html,2D3F888E8DCB501C020C3857C6E2A1E4 host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
			5:59:00.000 PM	host = BlueCoatProxy01 : source = /home/codepath/Files/Splunk-5-6-7/BlueCoatProxy01.csv : sourcetype = csv
>			6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3639.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv
>			6/4/23 5:59:00.000 PM	6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3639.218,200,EvilScript.exe,3AADB7E527FC1A050E1C97FEA1CBA4D host = uploadedhashes : source = /home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv : sourcetype = csv

6/4/23
6:00:00.000 PM

6/4/2023 18:00,192.168.1.8,EChan,23.21.189.88,http://twitch.tv,GET,Safari/13.1.1,200,2048,8192,Streaming Media

6/4/23
5:59:00.000 PM

6/4/2023 17:59,192.168.1.10,File Uploaded,Opera/75.0.3969.218,200,EvilScript.exe,3AADBF7E527FC1A050E1C97FEA1CB

Event Actions ▾

Type	☒ Field	Value	Actions
Selected	☒ host ▾	uploadedhashes	▾
	☒ source ▾	/home/codepath/Files/Splunk-5-6-7/uploadedhashes.csv	▾
	☒ sourcetype ▾	csv	▾
Event	☐ Event ▾	File Uploaded	▾
	☐ File Hash ▾	3AADBF7E527FC1A050E1C97FEA1CBA4D	▾
	☐ Filename ▾	EvilScript.exe	▾
	☐ IP ▾	192.168.1.10	▾
	☐ Response ▾	200	▾
	☐ Timestamp ▾	6/4/2023 17:59	▾
	☐ User Agent ▾	Opera/75.0.3969.218	▾
Time ⛶	_time ▾	2023-06-04T17:59:00.000-04:00	
Default	☐ index ▾	pathcode	▾
	☐ linecount ▾	1	▾

Challenge 14: Did any other users also upload a file around that time? If so, who and what was their IP address?

Solution:

- Yes, **Jmann**, uploaded a file on a different server using the ip: **(192.168.1.7)** at 15:29, while ABruke, who was the original uploader, uploaded on the main server his file at 15:21.

2. index=pathcode | search upload

1
index=pathcode
|
search
upload

✓ 4 events (before 10/21/24 7:35:33.000 PM)
No Event Sampling ▼

Events (4)
Patterns
Statistics
Visualization

Format Timeline ▼
Zoom Out
Zoom to Selection
Deselect

List ▼
Format
20 Per Page ▼

< Hide Fields
All Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
date_hour 1
date_mday 1
date_minute 2
a date_month 1
a date_wday 1
date_year 1

i	Time	Event
>	6/4/23 3:29:00.000 PM	6/4/2023 15:29,Jmann,192.168.1.7,File Upload,Firefox/89.0,200 host = webserver02 source = /home/codepath/Files/Splunk-5-6-7/webse
>	6/4/23 3:29:00.000 PM	6/4/2023 15:29,Jmann,192.168.1.7,File Upload,Firefox/89.0,200 host = webserver02 source = /home/codepath/Files/Splunk-5-6-7/webse
>	6/4/23 3:21:00.000 PM	6/4/2023 15:21,ABurke,192.168.1.10,File Upload,Firefox/89.0,200 host = webserver02 source = /home/codepath/Files/Splunk-5-6-7/webse
>	6/4/23 3:21:00.000 PM	6/4/2023 15:21,ABurke,192.168.1.10,File Upload,Firefox/89.0,200 host = webserver02 source = /home/codepath/Files/Splunk-5-6-7/webse

Username

2 Values, 100% of events
Selected

Reports

Top values
Top values by time
Rare values

Events with this field

Values	Count	%
ABurke	2	50%
Jmann	2	50%

Challenge 15: Looking at the uploaded hashes, what were the files called that the two users uploaded? Which one seems like it was malicious?

Solution:

- The two files uploaded by the two users were: (**proposal.pdf** by Jmann on (192.168.1.7) and **EvilScript.exe** by Aburke on (192.168.1.10)).

- The file **EvilScript.exe** uploaded by ABurke definitely seems like it was Malicious, because it is an executable, and two it is named "EVIL SCRIPT" nothing good can ever come from that!

host="uploadedhashes" Filename="*" (IP="192.168.1.7" OR "192.168.1.10") Event="File Uploaded"
| table Filename, Event, IP, _time

New Search

1 host="uploadedhashes" Filename="*" (IP="192.168.1.7" OR "192.168.1.10") Event="File Uploaded"

2 | table Filename, Event, IP, _time

✓ 4 events (before 10/21/24 9:41:36.000 PM) No Event Sampling

Events

Patterns

Statistics (4)

Visualization

20 Per Page Format Preview

Filename	Event	IP	_time
EvilScript.exe	File Uploaded	192.168.1.10	2023-06-04 17:59:00
proposal.pdf	File Uploaded	192.168.1.7	2023-06-04 15:30:00
proposal.pdf	File Uploaded	192.168.1.7	2023-06-04 15:30:00
EvilScript.exe	File Uploaded	192.168.1.10	2023-06-04 17:59:00

20 Per Page Format

Filename

EvilScript.exe

proposal.pdf

proposal.pdf

EvilScript.exe

Submission Checklist

👉 Check off each of the features you have completed. **You will only be graded on the features you check off.**

Reflection

- ☒ Reflection Question #1 answered above
- ☒ Reflection Question #2 answered above

CTF Challenges (10pts needed for full credit, 17pts needed for extra credit)

Part 1 – 1pt each

- ☒ Challenge #1: How many TV shows on Netflix are in the Docuseries genre?
- ☒ Challenge #2: How many movies on Netflix have a rating of TV-PG?
- ☒ Challenge #3: How many movies on Netflix were released in the year 2020?
- ☒ Challenge #4: What is the longest duration by season on Netflix, and what is its TV rating?
- ☒ Challenge #5: How many movies on Netflix are listed as action and are rated PG-13?
- ☒ Challenge #6: How many movies and TV shows on Netflix have their country of origin as Turkey?
- ☒ Challenge #7: Which release year had the most movies rated G? (Not TV-G)
- ☒ Challenge #8: What two TV-Y7 rated shows were released in 2019 and were added to Netflix on November 22, 2019?
- ☒ Challenge #9: Which year had the most movies from the United States?
- ☒ Challenge #10: What is the oldest TV show by Release Year on Netflix?

Part 2 – 2pts each

- ☒ Challenge #11: What was the IP address that uploaded the malware (MD5 hash: 3AADB7E527FC1A050E1C97FEA1CBA4D)?
- ☒ Challenge #12: What usernames did that IP address try to login to the system as? Which one did they upload a file as?
- ☒ Challenge #13: What was the User Agent String of the attacker when they successfully uploaded a file?
- ☒ Challenge #14: Did any other users also upload a file around that time? If so, who and what was their IP address?
- ☒ Challenge #15: Looking at the uploaded hashes, what were the files called that the two users uploaded? Which one seems like it was malicious?

 **Tip:** You can see specific grading information, including points breakdown, by going to [the grading page](#) on the course portal. 

Submit your work!

Step 1: **Click** the Share button at the top of your screen double check that anyone with the link can edit.



General access

 **Anyone with the link** ▼
Anyone on the internet with the link can edit

Editor ▼

Step 2: **Copy** the link to this document.

 [Copy link](#)

Step 3: **Submit** the link on the portal.